

```
1 package designpatternsandprinciple.ques6;
2
3 public interface Image {
4     void display();
5 }
```

```
1 package designpatternsandprinciple.ques6;
2
3 public class RealImage implements Image {
4
5     private String fileName;
6
7     public RealImage(String fileName) {
8         this.fileName = fileName;
9         loadFromServer();
10    }
11
12    private void loadFromServer() {
13        System.out.println("Loading image from remote server: " + fileName);
14    }
15
16    @Override
17    public void display() {
18        System.out.println("Displaying image: " + fileName);
19    }
20 }
21
```

```
2
3 public class ProxyImage implements Image {
4
5     private RealImage realImage;
6     private String fileName;
7
8     public ProxyImage(String fileName) {
9         this.fileName = fileName;
10    }
11
12    @Override
13    public void display() {
14
15        // Lazy Initialization
16        if (realImage == null) {
17            realImage = new RealImage(fileName);
18        }
19
20        realImage.display();
21    }
22 }
```

```
1 package designpatternsandprinciple.ques6;
2
3 public class ProxyTest {
4
5     public static void main(String[] args) {
6
7         Image image = new ProxyImage("nature.jpg");
8
9         System.out.println("First Display:");
10        image.display();
11
12        System.out.println();
13
14        System.out.println("Second Display:");
15        image.display();
16    }
17 }
18
```

First Display:
Loading image from remote server: nature.jpg
Displaying image: nature.jpg

Second Display:
Displaying image: nature.jpg